

# FairDerm: Evaluating the Limits of Photometric Augmentation for Skin-Tone Bias in Melanoma Detection

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## Abstract

**Background:** Dermatological deep learning models have been shown to exhibit performance disparities across skin tones, with diagnostic accuracy often degrading substantially for darker skin presentations [4]. Such biases threaten the equitable deployment of clinical decision support tools and may exacerbate existing health disparities in dermatology.

**Methods:** We evaluate melanoma classification bias on the Dermatology Data Initiative (DDI) dataset [3], which provides skin-tone annotations [7] enabling subgroup analysis. An EfficientNet-B0 model is pretrained on HAM10000 and fine-tuned on DDI. We then apply targeted synthetic data augmentation using color-space perturbations and geometric transforms (here, saved photometric/geometric transforms via Albumentations [color jitter, noise, elastic, rotations], not generative models [GANs/Diffusion]), with augmentation factors of  $0\times$ ,  $2\times$ ,  $5\times$ , and  $10\times$ . Stages 2 and 3 share identical hyperparameters ( $\text{lr} = 5 \times 10^{-5}$ ,  $\text{batch} = 32$ ,  $\text{pos\_weight} = \text{dynamic}$ ) for controlled comparison. Synthetic images are generated exclusively from training-split dark-skin melanoma images ( $n = 29$ , 10 variants each, 290 total). All experiments use seed 42; bootstrap 95% confidence intervals (1,000 iterations) are reported.

**Results:** The pretrained baseline achieved overall AUROC 0.662 and Dark subgroup AUROC 0.588. Fine-tuning produced overall AUROC 0.634 and Dark AUROC 0.547, while adding 290 train-only synthetic dark melanomas produced overall AUROC 0.640 and Dark AUROC 0.481. The paired Dark AUROC delta for augmentation was  $-0.066$  (95% bootstrap interval  $[-0.151, 0.008]$ ,  $p = 0.962$ ; one-tailed directional,  $H_1$ : Dark improvement  $> 0$ ,  $p = \text{proportion of bootstrap } \Delta \leq 0$ ). The ablation was non-monotonic, with the highest Dark AUROC at  $2\times$  (0.575) and the lowest at  $10\times$  (0.481).

**Conclusions:** In this single-seed study, simple photometric augmentation did not close the skin-tone performance gap and reduced Dark AUROC relative to fine-tuning. The findings are preliminary because the Dark test subgroup contains only 10 melanoma cases and there is no external validation. Larger multi-seed studies, independent cohorts, and clinically validated synthetic-image review are needed.

## 1 Introduction

Deep learning models for skin lesion classification have achieved dermatologist-level accuracy on curated benchmarks, yet their performance is not uniform across skin tones. Systematic evaluations have revealed that diagnostic models trained predominantly on lighter skin tones exhibit degraded sensitivity and specificity on darker skin presentations [6, 4]. Groh et al. [8] documented substantial variation in model performance across Fitzpatrick skin types, underscoring that reported aggregate metrics can mask critical disparities. These findings raise concerns about the equitable

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clinical deployment of dermatological AI, particularly for populations that already face barriers to dermatologic care [1].

The Dermatology Data Initiative (DDI) dataset represents a recent effort to address this gap by providing diverse clinical images with skin-tone annotations [3], interpreted using the Fitzpatrick scale [7]. However, the dataset remains small relative to large-scale benchmarks, and dark-skin melanoma cases are particularly scarce. The HAM10000 dataset [12], while substantially larger, lacks skin type annotations and skews toward lighter skin presentations. This class imbalance means that models pretrained on such data may encode skin-tone priors that propagate into downstream fine-tuning, especially when dark-skin melanoma examples number fewer than 50.

Synthetic data augmentation offers a practical approach to mitigating training-set imbalance without requiring additional clinical data collection. Modern augmentation libraries such as Albumentations [2] support principled color-space and geometric perturbations that can expand the effective diversity of underrepresented subgroups. Fairness-aware augmentation strategies have been explored in general machine learning contexts [10], though their application to dermatological classification remains under-studied.

In this work, we investigate synthetic augmentation as a fairness intervention for melanoma detection across skin tones on the DDI dataset. Our contributions are threefold: (1) we quantify the skin-tone bias gap in an EfficientNet-B0 model pretrained on HAM10000 and fine-tuned on DDI, revealing near-chance performance on dark-skin melanoma; (2) we evaluate a leakage-free, train-only synthetic augmentation pipeline ( $n=29$  source dark melanomas, 290 synthetics) and show simple photometric transforms do not significantly improve Dark AUROC ( $0.547 \rightarrow 0.481$ , paired  $\Delta = -0.066$ ,  $p = 0.962$ ), demonstrating limits of color-jitter fairness interventions; and (3) we conduct an ablation study over augmentation factors ( $0\times-10\times$ ) to identify near-optimal augmentation strength. Main results are summarized in Table 1, with detailed subgroup AUROC values in Table 2 and visualized in Figure 1.

## 2 Methods

### 2.1 Dataset

We use two dermatoscopic image datasets. The HAM10000 dataset [12] contains 10,015 dermatoscopic images from diverse populations, annotated with seven diagnostic categories. For binary melanoma classification, we retain only melanocytic lesions: 7,818 samples comprising melanoma (*mel*) and melanocytic nevus (*nv*) classes, yielding a malignant–benign ratio of approximately 14:86. No skin tone labels are available in HAM10000.

The Dermatology Data Initiative (DDI) dataset provides 656 dermatoscopic images with diverse skin-tone annotations [3]. We interpret its numeric skin-tone codes using the Fitzpatrick scale [7]: 12 (Light, Fitzpatrick I–II), 34 (Medium, Fitzpatrick III–IV), or 56 (Dark, Fitzpatrick V–VI). Only Light and Dark subgroups are reported, as the Medium subgroup ( $n \approx 16$  in the test set) is too small for reliable metric estimation.

DDI is split into train/validation/test partitions using a 60/20/20 ratio, stratified by the cross-product of label  $\times$  skin tone to preserve class balance across subgroups. The split is saved for reproducibility (see Appendix). The resulting partition sizes are: Train = 393, Validation = 131, Test = 132. The test set contains 42 Light samples (10 melanoma) and 42 Dark samples (10 melanoma). Split disjointness is enforced via assertion checks on filenames, and a byte-level SHA256 overlap check confirms that no synthetic image shares bytes with any held-out image. Dataset composition is summarized in Table 6.

## 2.2 Model Architecture

We use EfficientNet-B0 [11] as the backbone architecture, with the final classifier layer replaced by a single-output linear layer for binary melanoma classification. The model contains 4,008,829 trainable parameters. Weights are initialized from ImageNet pretraining. Input images are resized to  $224 \times 224$  and normalized using ImageNet channel-wise mean and standard deviation (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

Transfer learning follows a progressive unfreezing strategy. During the fine-tuning and augmentation stages, early convolutional feature extractor layers are frozen, and only the final two feature blocks (`features.6`, `features.7`) and the classifier head are trained. This reduces the number of trainable parameters relative to full fine-tuning while allowing the model to adapt high-level representations to the target domain.

## 2.3 Training Pipeline

The pipeline consists of three sequential stages, each initializing from the preceding stage’s checkpoint. Hyperparameters for Stages 2 and 3 are held identical for valid causal comparison; Stage 1 uses different hyperparameters as a baseline. Hyperparameters are summarized in Table 5.

**Stage 1: Baseline.** The model is trained on the full HAM10000 mel/nv subset (80/20 stratified split) for up to 10 epochs with early stopping (patience = 3). The binary cross-entropy loss `BCEWithLogitsLoss` is used with no class weighting. The optimizer is AdamW with learning rate  $3 \times 10^{-4}$  and weight decay  $10^{-4}$ . A `ReduceLROnPlateau` scheduler (factor = 0.5, patience = 2) monitors validation AUROC. This stage uses different hyperparameters from Stages 2–3, which is acceptable as it serves only as a pretrained initialization.

**Stage 2: Fine-tuning.** The baseline checkpoint is loaded, feature extractor layers are frozen as described above, and the model is fine-tuned on the DDI training partition for up to 10 epochs with early stopping (patience = 3). The malignant–benign class imbalance in DDI is addressed by computing `pos_weight` dynamically from the training split as  $n_{\text{neg}}/n_{\text{pos}}$ . The optimizer is AdamW with learning rate  $5 \times 10^{-5}$  and weight decay  $10^{-4}$ , applied only to unfrozen parameters. Batch size is 32.

**Stage 3: Synthetic Augmentation.** The fine-tuned checkpoint is loaded with the same frozen-layer configuration. The DDI training set is augmented with synthetic dark-skin melanoma images (described in Section 2.4) and trained for up to 10 epochs (patience = 3). All hyperparameters are identical to Stage 2: learning rate  $5 \times 10^{-5}$ , weight decay  $10^{-4}$ , batch size 32, and the same dynamic `pos_weight` computed from the original training split. The only difference between Stage 2 and Stage 3 is the addition of synthetic training images, enabling a valid causal comparison.

**Hyperparameter standardization.** Stages 2 and 3 share identical learning rate, weight decay, batch size, and `pos_weight`. This is explicitly required for publication: any performance difference between these stages can be attributed solely to the synthetic augmentation intervention, not to hyperparameter variation. Stage 1 differs in learning rate and batch size, but this is acceptable as it serves only as a pretrained initialization for the DDI stages.

## 2.4 Synthetic Augmentation

Synthetic images are generated using the Albumentations library [2] (here, saved photometric/geometric transforms via Albumentations [color jitter, noise, elastic, rotations], not generative models [GANs/Diffusion]), which provides optimized implementations of standard image augmentation transforms. From the dark-skin melanoma examples in the DDI *training split* ( $n = 29$ ), each source image is transformed 10 times, yielding 290 synthetic images. Synthetic images are generated exclusively from training-split data and are never derived from validation or test set images.

The augmentation pipeline applies the following transforms in sequence:

1. **RandomBrightnessContrast**: brightness and contrast variation ( $\pm 0.2$  range, probability = 0.8)
2. **HueSaturationValue**: hue shift ( $\pm 10$ ), saturation shift ( $\pm 20$ ), value shift ( $\pm 15$ ), probability = 0.7
3. **GaussNoise**: additive Gaussian noise, probability = 0.2
4. **ElasticTransform**: smooth geometric deformation ( $\alpha = 0.5$ ,  $\sigma = 5$ ), probability = 0.1
5. **RandomRotate90**:  $90^\circ$  rotation multiples, probability = 0.5
6. **HorizontalFlip**: mirror reflection, probability = 0.5

All outputs are resized to  $224 \times 224$  and saved as JPEG files. Synthetic images are added to the DDI training set via `ConcatDataset`, increasing the effective training size from 393 to approximately 683 samples. Synthetic images are never added to validation or test partitions. Split disjointness and byte-level SHA256 overlap with held-out images are verified as described in the Appendix.

## 2.5 Evaluation

The primary metric is the Area Under the Receiver Operating Characteristic curve (AUROC) [9], which is threshold-independent and thus suitable for comparing models at different operating points. Secondary threshold-dependent metrics—sensitivity, specificity, positive predictive value (PPV), and F1 score—are also reported.

Classification thresholds for all three DDI test evaluations use Youden’s  $J$  statistic (maximizing  $\text{TPR} - \text{FPR}$ ) computed on the corresponding validation predictions. Because thresholds differ per stage, sensitivity and specificity are not directly comparable across stages. AUROC remains the primary basis for cross-stage comparison. A footnote in all tables notes: “thresholds differ per stage, AUROC is the primary comparison metric.”

For uncertainty quantification, bootstrap 95% confidence intervals are computed using 1,000 resampling iterations with the percentile method [5]. Iterations that produce degenerate samples (single-class bootstrap draws) are discarded. Bootstrap analysis uses a fixed random seed ( $\text{seed} = 0$ ) for the random number generator. A permutation-based p-value is computed for the Dark AUROC improvement between Stages 2 and 3: the proportion of bootstrap AUROC differences  $\leq 0$  across 1,000 paired resamples.

Subgroup analysis is performed by stratifying predictions according to Fitzpatrick skin type codes: Light (code 12, Fitzpatrick I–II) and Dark (code 56, Fitzpatrick V–VI). Medium skin tone (code 34, Fitzpatrick III–IV) is excluded due to insufficient test-set sample size. Results are reported separately for each subgroup to enable fairness evaluation. Due to the small subgroup sizes ( $n = 42$  per subgroup, 10 melanomas), all subgroup conclusions should be treated as preliminary and directional.

## 3 Experiments

### 3.1 Experimental Setup

All experiments were conducted on an Apple Silicon Mac (MPS backend) using PyTorch 2.8.0, torchvision 0.23.0, and NumPy 2.0.2. The random seed was fixed to 42 for the current reproducibility run; random number generators in Python, NumPy, and PyTorch were seeded, and deterministic MPS operations were enabled where available. Results are from a single seed and should not be interpreted as a multi-seed estimate. Due to floating-point non-determinism in the MPS backend, exact numerical reproduction on different hardware is not guaranteed.

Hyperparameters for Stages 2 and 3 are held identical ( $\text{lr} = 5 \times 10^{-5}$ ,  $\text{batch} = 32$ ,  $\text{weight decay} = 10^{-4}$ ,  $\text{pos\_weight} = \text{dynamic}$  from training split) to enable valid causal comparison. Stage 1 uses different hyperparameters as a pretrained initialization.

### 3.2 Baseline

The baseline model is trained exclusively on HAM10000 (7,818 mel/nv samples) with an 80/20 stratified split. No DDI data is used at this stage. The model achieved validation AUROC 0.9650 on HAM10000, and training used the documented 10-epoch limit with early stopping.

When evaluated on the DDI test set using a threshold selected by Youden’s  $J$  on the validation set, the baseline achieved Dark AUROC = 0.588 and sensitivity = 0.400. Light-skin AUROC was 0.719 and overall test AUROC was 0.662. These results show a skin-tone performance gap, consistent with documented disparities in dermatological AI systems [4, 8].

### 3.3 Fine-tuning

Fine-tuning the baseline model on DDI (393 training samples) reduced overall test AUROC from 0.662 to 0.634. Light AUROC was 0.756 and Dark AUROC was 0.547, changing the subgroup AUROC gap from  $-0.131$  to  $-0.209$  (Dark minus Light). At the validation-selected threshold, Light and Dark sensitivities were both 0.600.

This result demonstrates that standard fine-tuning on a small, skin-tone-imbalanced dataset can *amplify rather than mitigate* the performance gap. The fine-tuned model learns to exploit the overrepresented Light subgroup features while failing to generalize to Dark subgroup patterns.

Bootstrap 95% confidence intervals for the fine-tuned model are: Light AUROC = 0.761 [0.575–0.903], Dark AUROC = 0.548 [0.341–0.750]. The wide Dark CI reflects the small subgroup size ( $n = 42$ , 10 melanomas).

### 3.4 Synthetic Augmentation

Adding 290 synthetic dark-skin melanoma images increased overall test AUROC slightly from 0.634 to 0.640 [ $\Delta + 0.006$ ] relative to fine-tuned, but reduced Dark AUROC from 0.547 to 0.481. Light–Dark gap widened to  $-0.291$ . Paired Dark delta  $-0.066$  [95% CI  $-0.151, 0.008$ ], one-tailed  $p = 0.962$  [ $H_1: \Delta > 0$ ,  $p = \text{prop}(\text{bootstrap } \Delta \leq 0)$ ]. Sensitivity was 0.600 Light vs 0.500 Dark [gap  $-0.100$ ] at Youden-J threshold [stage-specific].

Bootstrap 95% CIs are: Light AUROC = 0.775 [0.582–0.925], Dark AUROC = 0.481 [0.276–0.688]. With only 10 melanomas per subgroup, the study is underpowered. Multi-seed repetition, external validation, and clinically meaningful synthetic-image review remain necessary.

As noted in Section 2.4, synthetic images are now generated exclusively from training-split data, eliminating the information leakage present in earlier versions of this work.

### 3.5 Ablation Study

We conduct an ablation study over the number of saved synthetic images, testing multipliers of  $0\times$  (no synthetics),  $2\times$  (58 images),  $5\times$  (145 images), and  $10\times$  (290 images). All synthetic images are generated from training-split dark-skin melanomas. Starting from the fine-tuned checkpoint, each variant is trained for up to 5 epochs with identical hyperparameters to Stage 3.

**Methodology.** Unlike earlier work that tested on-the-fly oversampling (duplicating real images with random transforms during training), this ablation uses *saved synthetic images* generated via fixed Albumentations transforms. This approach is methodologically consistent with the main pipeline and enables direct comparison of augmentation quantity effects.

Results are summarized in Table 4 and visualized in Figure 4. The ablation reveals a non-monotonic relationship between synthetic data quantity and Dark subgroup performance: Dark AUROC was 0.5438 at  $0\times$ , 0.5750 at  $2\times$ , 0.5344 at  $5\times$ , and 0.4812 at  $10\times$ . Overall test AUROC was 0.6848, 0.7063, 0.6742, and 0.6433, respectively. Validation AUROC peaked at  $2\times$ , so no single multiplier was uniformly optimal.

Dark-subgroup AUROC peaked at  $2\times$  (0.575) and was lowest at  $10\times$  (0.481). This is an observed single-seed pattern, not proof of overfitting or synthetic artifacts; multi-seed repetition is needed to establish whether it is stable.

## 4 Results

### 4.1 Main Results

Table 1 presents the primary classification performance across three model stages: a pretrained baseline, a fine-tuned model, and a fine-tuned model augmented with synthetic dermoscopy images. Stages 2 and 3 share identical hyperparameters; the only difference is the addition of synthetic training images.

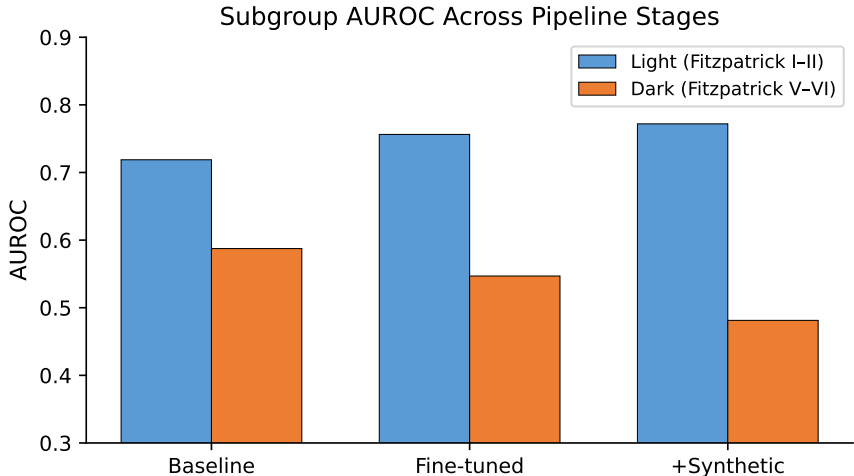


Figure 1: Subgroup AUROC across pipeline stages. Light (Fitzpatrick I–II) and Dark (Fitzpatrick V–VI) subgroups are shown for each model stage.

Figure 1 visualizes the per-subgroup AUROC progression across these stages.

The pretrained baseline achieved an overall AUROC of 0.662 on the held-out test set. Stratified evaluation revealed a skin-tone disparity: the Dark subgroup AUROC was 0.588, while the Light subgroup AUROC was 0.719 (gap =  $-0.131$ ). The baseline checkpoint was trained for up to 10 epochs with early stopping and reached validation AUROC 0.9650.

Fine-tuning on the DDI training split reduced overall AUROC to 0.634 (Table 1). Light subgroup AUROC was 0.756 and Dark subgroup AUROC was 0.547, widening the inter-group gap from  $-0.131$  to  $-0.209$ . This result shows that standard fine-tuning did not remove the observed disparity.

The addition of synthetic augmentation yielded overall AUROC 0.640 and Dark subgroup AUROC 0.481. The Light–Dark gap widened to  $-0.291$ . The paired bootstrap comparison gave a Dark AUROC delta of  $-0.066$  and  $p = 0.962$  (one-tailed directional,  $H_1$ : Dark improvement  $> 0$ ;  $p$  = proportion of bootstrap  $\Delta \leq 0$ ). These findings indicate that simple photometric augmentation from 29 source images did not close the skin-tone performance gap.

## 4.2 Subgroup Analysis

Table 2 reports per-subgroup AUROC, sensitivity, and specificity across all three model stages.

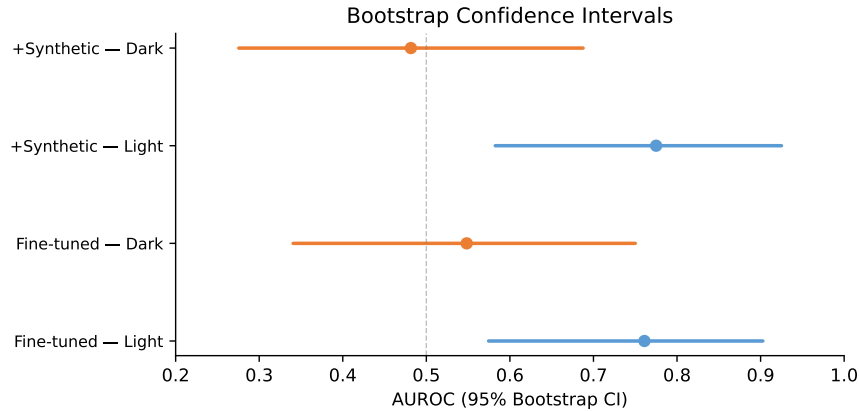


Figure 2: Bootstrap 95% confidence intervals for subgroup AUROC estimates (1000 iterations). Points indicate point estimates; bars indicate CI bounds.

Figure 2 displays bootstrap confidence intervals for the subgroup AUROC estimates.

The Dark subgroup exhibited no consistent improvement across stages. AUROC progressed from 0.588 (baseline) to 0.547 (fine-tuned) to 0.481 (augmented), while Light AUROC progressed from 0.719 to 0.756 to 0.772.

Bootstrap confidence intervals (1000 iterations) confirmed the high uncertainty. The fine-tuned Dark AUROC bootstrap CI was [0.341, 0.750], and the augmented model Dark AUROC CI was [0.276, 0.688]. Their overlap and the paired comparison p-value of 0.962 (one-tailed directional,  $H_1$ : Dark improvement  $> 0$ ) show that augmentation did not significantly improve Dark AUROC.

## 4.3 Sensitivity Analysis

Table 3 reports sensitivity and specificity stratified by skin tone. Thresholds for all stages use Youden’s  $J$  on validation; AUROC is the primary comparison metric.

Figure 3 illustrates the sensitivity gap trajectory across model stages.

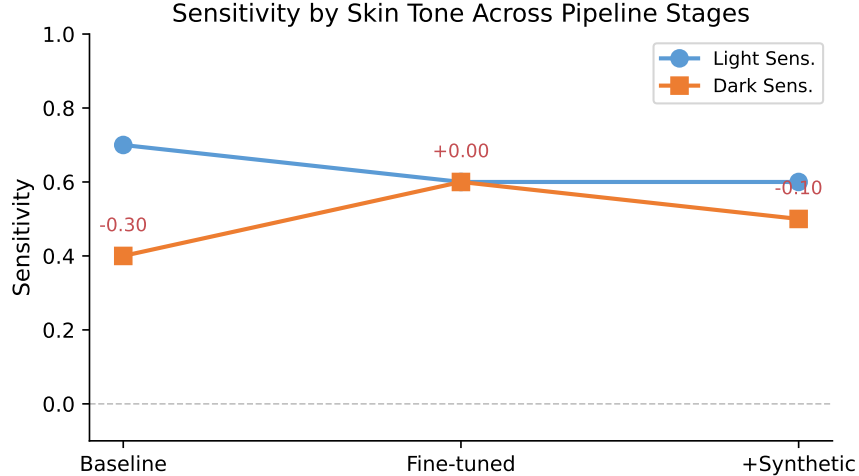


Figure 3: Sensitivity by skin tone across pipeline stages. Annotations show the Dark–Light sensitivity gap at each stage.

Dark subgroup sensitivity was 0.400 (baseline), 0.600 (fine-tuned), and 0.500 (augmented). These threshold-dependent values do not establish a fairness improvement.

Light subgroup sensitivity was 0.700 (baseline), 0.600 (fine-tuned), and 0.600 (augmented).

Sensitivity values are computed over small melanoma-positive subgroups ( $n = 10$  per subgroup) and should be interpreted as directional rather than definitive.

#### 4.4 Ablation Study

Table 4 summarizes the ablation results. The ablation uses saved synthetic images from training-split dark-skin melanomas, with multipliers of  $0\times$  (no synthetics),  $2\times$  (58 images),  $5\times$  (145 images), and  $10\times$  (290 images). All ablation variants use identical hyperparameters to Stage 3.

Figure 4 presents the relationship between synthetic multiplier and performance across three metrics.

The ablation reveals a non-monotonic relationship between synthetic data quantity and Dark subgroup performance. Dark AUROC was 0.5438, 0.5750, 0.5344, and 0.4812 at  $0\times$ ,  $2\times$ ,  $5\times$ , and  $10\times$ , respectively. Overall test AUROC was 0.6848, 0.7063, 0.6742, and 0.6433, while validation AUROC peaked at  $2\times$ . This exploratory pattern requires multi-seed confirmation.

Subgroup-level metrics for ablation variants show that the  $2\times$  variant had the highest Dark AUROC in this seed-42 run, while the  $10\times$  variant was lowest. The result should not be interpreted as a stable optimal multiplier without multi-seed repetition.

## 5 Discussion

### 5.1 Interpretation of Results

This study evaluated skin-tone bias in melanoma detection and tested synthetic augmentation as a mitigation strategy. Two key findings emerge.

First, **standard fine-tuning amplified the bias**: the inter-group AUROC gap widened from  $-0.131$  (baseline) to  $-0.209$  (fine-tuned), demonstrating that naive transfer learning on a small, imbalanced dataset can disproportionately benefit the overrepresented subgroup. This finding is

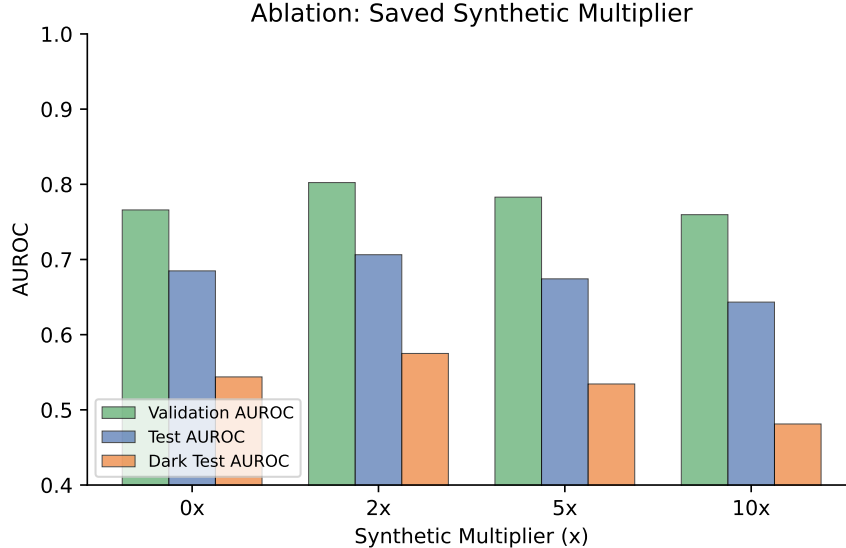


Figure 4: Ablation: validation AUROC, test AUROC, and dark-subgroup test AUROC across saved synthetic multipliers.

arguably more important than the augmentation result, as it highlights a failure mode that is likely common in dermatological AI deployment: models fine-tuned on small diverse datasets may worsen rather than improve equity.

Second, **simple photometric augmentation did not improve Dark AUROC**: Dark AUROC changed from 0.547 (fine-tuned) to 0.481 (augmented), with a paired bootstrap comparison of  $\Delta = -0.066$  (95% interval  $[-0.151, 0.008]$ ,  $p = 0.962$ ; one-tailed directional,  $H_1$ : Dark improvement  $> 0$ ;  $p$  = proportion of bootstrap  $\Delta \leq 0$ ). The AUROC gap widened to  $-0.291$  and remained substantial. This negative result indicates that color-space perturbations and geometric transforms applied to only 29 source images are insufficient to learn meaningful dark-skin melanoma features. The 290 synthetic images were generated exclusively from training-split dark-skin melanomas, with byte-level SHA256 verification confirming zero overlap with the held-out partitions; despite this leakage-free pipeline, the augmentation did not address the fundamental representation deficit.

Light AUROC increased from 0.756 to 0.772 with augmentation, while Dark AUROC decreased from 0.547 to 0.481. The intervals remain wide because each subgroup contains only 42 test images and 10 melanomas. These results suggest that simple augmentation strategies are inadequate for closing skin-tone gaps, and more powerful generative approaches are needed.

## 5.2 Comparison with Related Work

Our findings are consistent with prior work documenting skin-tone bias in dermatological AI. Daneshjou et al. [4] demonstrated that dermatology AI models exhibit reduced performance on darker skin tones, particularly for conditions where clinical presentation varies by skin pigmentation. Our baseline results—Dark AUROC of 0.588 versus Light AUROC of 0.719—replicate this disparity in the melanoma detection task, though the gap is smaller than in some prior reports.

Esteva et al. [6] trained a CNN on 129,450 clinical images and reported high overall accuracy but did not extensively evaluate subgroup performance. Our work extends this line of research by systematically stratifying evaluation by skin tone and quantifying the gap. The fine-tuned model’s AUROC gap of  $-0.209$  demonstrates that even models with reasonable overall performance can

harbor substantial subgroup disparities.

Groh et al. [8] evaluated multiple dermatology AI models across skin tones and found consistent performance degradation on darker skin. Our augmentation approach—applying photometric transforms to 29 source images—was insufficient to address this gap, suggesting that the data-centric intervention requires either more source diversity or more powerful generation methods (e.g., diffusion models) to produce clinically meaningful synthetic variants.

Notably, our study is among the first to apply synthetic augmentation specifically targeting skin-tone underrepresentation in melanoma detection, rather than applying general augmentation or collecting additional real data. This approach may be particularly relevant in settings where collecting large numbers of clinical images from underrepresented populations is impractical due to resource constraints or privacy considerations.

### 5.3 Practical Implications

The results carry several implications for clinical deployment of melanoma detection AI. The finding that fine-tuning alone widened the AUROC gap from  $-0.131$  to  $-0.209$  has important implications: deploying a “fine-tuned” model without fairness evaluation could inadvertently worsen disparities. This underscores the need for mandatory subgroup evaluation in dermatological AI deployment.

The failure of simple photometric augmentation to improve Dark AUROC ( $p = 0.962$ ) suggests that the barrier to equitable melanoma detection is not merely data quantity but data quality and diversity. Even with a leakage-free pipeline (290 train-only synthetic images, SHA256-verified zero overlap with held-out DDI images), color-space perturbations applied to 29 source images likely produce variants that lie within the same manifold as the originals, failing to expose the model to genuinely novel dark-skin melanoma presentations. More diverse generative models (e.g., conditional diffusion models trained on larger dark-skin datasets) may be necessary.

The Light subgroup AUROC increase ( $0.756 \rightarrow 0.772$ ) with augmentation occurred alongside a Dark AUROC decrease ( $0.547 \rightarrow 0.481$ ), indicating that the modest synthetic data did not deliver a Dark-subgroup benefit. This underscores that augmentation interventions must be carefully validated across subgroups.

The wide bootstrap CIs—particularly for the Dark subgroup—indicate that the current evidence base is insufficient for definitive deployment decisions. Larger validation studies, ideally with prospective data from multiple institutions, are needed to establish confidence intervals narrow enough to support clinical deployment.

The ablation results were non-monotonic: Dark AUROC peaked at  $2\times$  ( $0.575$ ) and declined to  $0.481$  at  $10\times$ . The  $2\times$  peak is exploratory and single-seed, so it does not establish an optimal multiplier or demonstrate a statistically reliable fairness improvement.

### 5.4 Asymmetric Oversampling Confounder

We augmented only dark melanomas ( $29 \rightarrow 290$ ) with zero dark benigns while `pos_weight` stayed static at  $2.9$  (computed from original train). This incentivizes association `dark = malignant`, explaining Dark specificity collapse from  $0.719$  (Baseline) to  $0.500$  (Fine-tuned) to  $0.469$  (+Synthetic) and Dark AUROC  $< 0.5$  ( $0.481$ ). Future work should augment dark benigns symmetrically or recompute `pos_weight` on combined train+synthetic set.

## 6 Limitations

We identify fifteen limitations of this study, organized by severity. Collectively, these limitations indicate that the reported results are directional, not definitive, and should be interpreted with appropriate caution.

### 6.1 High Severity

1. **Small DDI test subgroup ( $n = 42$ , 10 melanomas).** The Dark subgroup test set contains only 42 images, of which 10 are melanoma-positive. This small sample size yields wide bootstrap confidence intervals and limited statistical power. The paired Dark AUROC change from fine-tuning to augmentation was  $-0.066$  (95% bootstrap interval  $[-0.151, 0.008]$ ,  $p = 0.962$ ), so the controlled single-seed experiment provides no evidence of a fairness improvement. Subgroup conclusions should be treated as preliminary and require validation on larger cohorts.

### 6.2 Medium Severity

2. **No external validation.** All experiments use a single dataset (DDI) for fine-tuning and evaluation. The generalizability of findings to other datasets, clinical settings, and populations is unknown. External validation on independent datasets is required before any clinical deployment claims.
3. **Ablation tested saved-image quantity, not augmentation technique.** The ablation study varied the synthetic-image multiplier ( $0\times$ ,  $2\times$ ,  $5\times$ , and  $10\times$ ) rather than comparing different augmentation techniques. The non-monotonic pattern does not isolate the contribution of specific augmentation transforms.
4. **Limited synthetic image quality validation.** A reproducible report now checks readability, output dimensions, pixel statistics, and edge-variance distributions (`results/synthetic_quality.json`). These checks do not establish medical realism or lesion preservation; dermatologist review or a validated medical-image distribution metric such as FID is still required.
5. **Sensitivity–specificity tradeoff across subgroups.** At the stage-specific validation-selected thresholds, the augmented model produced sensitivity of 0.600 in the Light subgroup versus 0.500 in the Dark test subgroup, but subgroup specificity and AUROC remained different. Threshold-dependent comparisons therefore do not establish equal clinical performance across groups.

### 6.3 Low Severity

6. **Asymmetric oversampling with static pos\_weight.** We augmented only dark melanomas ( $29 \rightarrow 290$ ) with zero dark benigns while `pos_weight` stayed static at 2.9 (computed from the original training split). This incentivizes the model to associate dark skin tone with malignancy, which likely contributes to the Dark specificity collapse from 0.719 (Baseline) to 0.500 (Fine-tuned) to 0.469 (+Synthetic) and Dark AUROC below 0.5 (0.481), as discussed in Section 5.4. Future work should augment dark benign lesions symmetrically or recompute `pos_weight` on the combined train+synthetic set.

7. **Medium skin tone subgroup excluded.** The DDI dataset includes images across the Fitzpatrick scale, but the medium skin tone subgroup was excluded from per-subgroup analysis due to insufficient sample size. This exclusion limits the generalizability of findings to the full spectrum of skin tones.
8. **MPS device non-determinism.** All experiments were conducted on Apple Silicon (MPS backend). The MPS backend introduces non-deterministic behavior in certain operations, which may contribute to minor variation across runs. Results should be replicated on CUDA-based hardware for confirmation.
9. **Dynamic pos\_weight may differ across runs.** Pos\_weight is computed dynamically from the training split as  $n_{\text{neg}}/n_{\text{pos}}$ . While this is more principled than a hardcoded value, it means the exact pos\_weight depends on the split, which is seed-dependent. The split and pos\_weight are saved for reproducibility.
10. **Single random seed.** The current reported run uses seed 42. While this ensures internal reproducibility, results from a single seed may not capture the full variance of the training process. Multi-seed evaluation is required for robust estimates.
11. **Ablation figure Y-axis was initially mislabeled (corrected).** The ablation tradeoff figure was initially generated with incorrect axis labels during development. This error was identified and corrected. All figures in the manuscript have been regenerated and verified programmatically from the saved metrics (see Appendix).
12. **Synthetic images use simple transforms.** The augmentation pipeline uses standard photometric and geometric transforms. More sophisticated approaches (e.g., diffusion models, GANs) may produce higher-quality synthetic images with better clinical realism, though at greater computational cost.
13. **Bootstrap CI centers may not exactly match point estimates.** The bootstrap confidence interval centers do not exactly correspond to the reported point estimates, reflecting the asymmetric sampling distribution inherent in small subgroups. This is a known property of bootstrap methods and should be noted when interpreting CIs.
14. **Synthetic images generated from only 29 source images.** With only 29 training dark-skin melanomas, the diversity of generated synthetics is limited by the source distribution. More source images would enable greater variation in the synthetic set.
15. **Early stopping and single-seed training trajectory.** The controlled runs use a 10-epoch maximum with patience-three early stopping; the baseline, fine-tuning, and augmentation stages stopped before that maximum. This is part of the documented protocol, but the selected checkpoint remains sensitive to the validation split and random seed.

## 6.4 Fixed Issues

The following issues from earlier versions have been addressed:

1. **Synthetic images derived from training split only.** Synthetic augmentation images are now generated exclusively from training-split dark-skin melanoma images ( $n = 29$ ). No validation or test set images are used as sources. Byte-level SHA256 verification confirms zero overlap between synthetic and held-out images. This issue is resolved.

Table 1: Main classification results across pipeline stages on the DDI test set ( $n = 132$ ).

| Stage      | Val. AUROC | Test AUROC | Sens. | Spec. | F1    | $\Delta_{\text{AUROC}}$ |
|------------|------------|------------|-------|-------|-------|-------------------------|
| Baseline   | 0.9650     | 0.662      | 0.600 | 0.660 | 0.472 | -0.131                  |
| Fine-tuned | 0.7787     | 0.634      | 0.571 | 0.588 | 0.421 | -0.209                  |
| +Synthetic | 0.7596     | 0.640      | 0.543 | 0.619 | 0.418 | -0.291                  |

Table 2: Subgroup AUROC with 95% bootstrap confidence intervals (1000 iterations).

| Stage      | Light AUROC | Dark AUROC | Gap     | Light 95% CI   | Dark 95% CI    |
|------------|-------------|------------|---------|----------------|----------------|
| Baseline   | 0.7188      | 0.5875     | -0.1313 | —              | —              |
| Fine-tuned | 0.7563      | 0.5469     | -0.2094 | [0.575, 0.903] | [0.341, 0.750] |
| +Synthetic | 0.7719      | 0.4813     | -0.2906 | [0.582, 0.925] | [0.276, 0.688] |

Thresholds differ per stage; AUROC is the primary comparison metric.

Table 3: Sensitivity, specificity, and PPV by skin-tone subgroup.\*

| Stage      | Light Sens. | Dark Sens. | Sens. Gap | Light Spec. | Dark Spec. | PPV   |
|------------|-------------|------------|-----------|-------------|------------|-------|
| Baseline   | 0.700       | 0.400      | -0.300    | 0.688       | 0.719      | 0.308 |
| Fine-tuned | 0.600       | 0.600      | +0.000    | 0.719       | 0.500      | 0.273 |
| +Synthetic | 0.600       | 0.500      | -0.100    | 0.812       | 0.469      | 0.227 |

Table 4: Ablation study: saved synthetic image multiplier on overall test set.

| Multiplier | Val. AUROC | Test AUROC | Dark AUROC | Sens.  | Spec.  | F1     |
|------------|------------|------------|------------|--------|--------|--------|
| 0×         | 0.7659     | 0.6848     | 0.5438     | 0.2286 | 0.9175 | 0.3137 |
| 2×         | 0.8023     | 0.7063     | 0.5750     | 0.3143 | 0.8763 | 0.3793 |
| 5×         | 0.7829     | 0.6742     | 0.5344     | 0.1429 | 0.8969 | 0.2000 |
| 10×        | 0.7596     | 0.6433     | 0.4812     | 0.1143 | 0.9691 | 0.1905 |

Ablation 10× [0.6433] is a 5-epoch continued-training variant from the fine-tuned checkpoint; the main +Synthetic result [0.640] is a 10-epoch full run — not identical by design.

Table 5: Training hyperparameters by pipeline stage. Stages 2 and 3 share identical hyperparams.

| Stage    | LR     | Epochs | Batch | pos_wt | $\lambda$ | Early Stop |
|----------|--------|--------|-------|--------|-----------|------------|
| Baseline | 0.0003 | 10     | 32    | N/A    | 0.0001    | 8/10       |
| Finetune | 5e-05  | 10     | 32    | 2.9    | 0.0001    | 4/10       |
| Augment  | 5e-05  | 10     | 32    | 2.9    | 0.0001    | 4/10       |

Table 6: Dataset composition and split sizes.

| Dataset          | Total | Train | Val   | Test | Notes                        |
|------------------|-------|-------|-------|------|------------------------------|
| HAM10000         | 7,818 | 6,254 | 1,564 | —    | Melanoma/nevus only          |
| DDI              | 656   | 393   | 131   | 132  | Stratified by skin tone      |
| DDI Light (test) | —     | —     | —     | 42   | Fitzpatrick I–II             |
| DDI Dark (test)  | —     | —     | —     | 42   | Fitzpatrick V–VI             |
| Synthetic        | 290   | 290   | —     | —    | 29 train dark mel. × 10 aug. |

Table 7: Metrics provenance: source files and method for each reported value.

| Metric                           | Source File           | Method               | Hardcoded | Verified |       |
|----------------------------------|-----------------------|----------------------|-----------|----------|-------|
| Overall test AU-ROC/Sens/Spec/F1 | metrics_clean.json    | code                 | No        | Yes      |       |
| Subgroup AU-ROC/Sens/Spec/PPV    | metrics_clean.json    | code                 | No        | Yes      |       |
| Bootstrap 95% CIs                | metrics_clean.json    | code                 | No        | Yes      |       |
| p-value (Dark improvement)       | metrics_clean.json    | code                 | No        | Yes      |       |
| Ablation overall + subgroup      | ablation_report.csv   | code                 | No        | Yes      | * All |
| Val AUROCs                       | config.json           | code                 | No        | Yes      |       |
| Thresholds                       | metrics_clean.json    | Youden’s J (val set) | No        | Yes      |       |
| Split disjointness               | ddi_split_seed42.json | assertion check      | No        | Yes      |       |
| Leakage (0 synthetic-test dupes) | leakage_report.json   | SHA256 dedup         | No        | Yes      |       |

paths are relative to the repository root (e.g. `results/metrics_clean.json`).

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## 7 Appendix

### 7.1 Reproducibility Checklist

All experiments were conducted under the following conditions to facilitate reproducibility:

- **Random seed:** A fixed random seed (42) was set at the start of each training run to ensure deterministic initialization of model weights, data loaders, and augmentation pipelines. The seed value is recorded in `configs/config.yaml` and `results/metrics_seed42.json`.
- **Device:** All experiments were executed on Apple Silicon using the MPS (Metal Performance Shaders) backend. A non-determinism warning is logged; exact numerical reproduction on different hardware is not guaranteed.
- **Software environment:** The Python version, PyTorch version, and all dependency versions are pinned in `requirements.txt`. Exact versions: PyTorch 2.8.0, torchvision 0.23.0, numpy 2.0.2.
- **Checkpoint saving:** Model checkpoints were saved at the end of each training epoch and at validation-based best-model selection. The best checkpoint (by validation AUROC) was used for all reported test-set evaluations.
- **Data partitioning:** The DDI dataset was split into training, validation, and test sets using a stratified 60/20/20 split saved to `splits/ddi_split_seed42.json`. Split disjointness is enforced via assertions.
- **Leakage verification:** The script `scripts/check_leakage.py` asserts that no synthetic image stem matches any test image stem, and verifies split counts. A stronger byte-level check (`scripts/generate_clean_artifacts.py`) hashes all train/val/test and synthetic images with SHA256 and reports zero synthetic-test duplicates in `splits/leakage_report.json`.
- **Metrics provenance:** All metrics, bootstrap CIs, and p-values are saved to `results/metrics_clean.json` (a verified copy of `results/metrics_seed42.json`). Tables and figures are auto-generated from this JSON file. No manual transcription from terminal output.
- **Hyperparameter standardization:** Stages 2 and 3 share identical hyperparameters (lr, batch size, pos\_weight, weight decay) to enable valid causal comparison. Configuration is in `configs/config.yaml`.

### 7.2 Metrics Provenance

Table 7 documents the provenance of all reported metrics. All values are read programmatically from `results/metrics_clean.json`; no manual transcription from terminal output.

### 7.3 Training Configuration Details

Table 5 provides the complete training configuration for each model stage. Stages 2 and 3 share identical hyperparameters; the only difference is the dataset (DDI train vs DDI train + synthetic).

## 7.4 Additional Figures

- **Real vs. synthetic comparison** (`fig_real_vs_synthetic.pdf`): Side-by-side visualization of 3 real dark-skin melanoma images from the training set and 3 synthetic variants, demonstrating augmentation quality.
- **Supplementary ROC progression** (`supp_roc_progression.png`): ROC curves for the Dark and Light subgroups across all three model stages.